Cretaceous remagnetization in the Sylvester Allochthon: limits to post-105 Ma northward displacement of north-central British Columbia

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The Sylvester Allochthon of the Slide Mountain Terrane in northern British Columbia is a structurally interleaved assemblage of ocean-floor lithologies ranging in age from Late Devonian to Late Triassic. It is the most inboard of oceanic suspect terranes and rests as a vast klippe on miogeoclinal strata of the Cassiar Terrane. The Sylvester Allochton and the Cassiar Terrane lie west of the Northern Rocky Mountain Trench Fault. Both the Sylvester Allochthon and the Cassiar Terrane are intruded by mid-Cretaceous (105 Ma) granite of the Cassiar Batholith. Six oriented cores were collected at each of 12 sites in Guadalupian *Parafusulina*-bearing limestone of the Sylvester Allochthon at a location 4 km from the batholith. Isothermal remanent magnetization (IRM) acquisition and subsequent thermal demagnetization indicate that pyrrhotite is the dominant ferrimagnetic mineral. Least-squares line fitting to four thermal demagnetization steps between 150 and 310°C was used to determine the characteristic natural remanent magnetization (NRM) directions that fail the fold test at the 99.5% confidence level. We interpret these observations as indicating that the NRM is a thermoremanent or thermochemical remanent magnetism associated with intrusion of the Cassiar Batholith. The resulting paleomagnetic pole location is latitude = 75.7°N, longitude = 171.7°E, α_{95} = 8.5°. When compared with the mid-Cretaceous pole for cratonic North America, a small but significant clockwise rotation ($R \pm \Delta R = 23.9 \pm 18.1$ °) is evident, but poleward translation ($p \pm \Delta p = 5.3 \pm 9.2$°) is not significant at the 95% confidence level. The paleomagnetic results are consistent with geological evidence for moderate (700 km) northward transport of the Cassiar Terrane (along with the previously emplaced overlying Sylvester Allochthon) during mid-Cretaceous to Tertiary dextral transcurrent faulting.

L'allochtone de Sylvester du terrane de Slide Mountain dans le nord de la Colombie-Britannique est un assemblage structuralement intercalé de lithologies de fond océanique variant en âge du Dévonien tardif au Trias tardif. Parmi les terranes océaniques suspects, il est le plus intérieur sur le continent et il forme une importante klippe reposant sur les strates miogéoclinales du terrane de Cassiar. L'allochtone de Sylvester et le terrane de Cassiar affleurent à l'ouest de la faille de Sillon Nord des Rocheuses. Le granite du batholite de Cassiar, d'âge crétacé moyen (105 Ma), recoupe les roches de l'allocthone de Sylvester ainsi que celles du terrane de Cassiar. Six carottes orientées ont été prélevées à chacun des 12 sites forés dans le calcaire à *Parafusulina*, d'âge guadalupien, de l'allochtone de Sylvester à une distance éloignée de 4 km du batholite. L'aimantation rémanente isothermique acquise et sa désaimantation thermique subséquente révèlent que la pyrrhotite est le minéral ferrimagnétique dominant. Une droite de régression des moindres carrés pour une désaimantation thermique en quatre paliers, entre 150 et 310°C, a servi à déterminer les directions de l'aimantation rémanente naturelle qui défaillaient au test de plissement avec niveau de confiance de 99,5%. Nous interprétons ces observations comme une aimantation rémanente naturelle de type thermorémanent ou thermochimique associée à l'intrusion du batholite de Cassiar. La position calculée du pôle magnétique est latitude = 75,7°N, longitude = 171,7°E, α_{95} = 8,5°. La comparaison avec le pôle pour le craton de l'Amérique du Nord au Crétacé moyen indique une rotation, faible mais significative ($R \pm \Delta R = 23{,}9 \pm 18{,}1$°), cependant le glissement vers le pôle ($p \pm \Delta p = 5{,}3 \pm 9{,}2$°) est négligeable au niveau de confiance de 95%. Les données paléomagnétiques sont en accord avec les observations géologiques pour un déplacement modéré (700 km) vers le nord du terrane de Cassiar (avec la klippe de Sylvester sus-jacente) durant la période de formation des failles de décrochement dextres entre le Crétacé moyen en le Tertiaire.

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Introduction

The Sylvester Allochthon is a structurally interleaved assemblage of lithotectonic slices of uppermost oceanic crust lithologies: massive and pillow basalt, gabbro, banded radiolarian chert, argillite, carbonate, serpentinite, and ultramafic rocks ranging in age from Late Devonian through Triassic (Harms 1986*a*). The allochthon may represent considerable delamination and telescoping of a proto-Pacific ocean outboard of Paleozoic North America (Harms 1985, 1986*b*). The Sylvester Allochthon lies above a planar, subhorizontal fault as a vast klippe on displaced Paleozoic North American miogeoclinal carbonate and clastic strata of the Cassiar Terrane (Fig. 1). Ancestral North American units are exposed beneath and flanking the Sylvester Allochthon to both east and west. The basal Sylvester Fault, along with the overlying allochthon and the underlying strata of the Cassiar Terrane, is cross-cut by the Cassiar Batholith at several localities along the western flank of the Sylvester Allochton (Gabrielse 1963). The Cassiar Batholith is one of a suite of mid-Cretaceous granites characteristic of the Omineca Belt. It is more than 300 km long and

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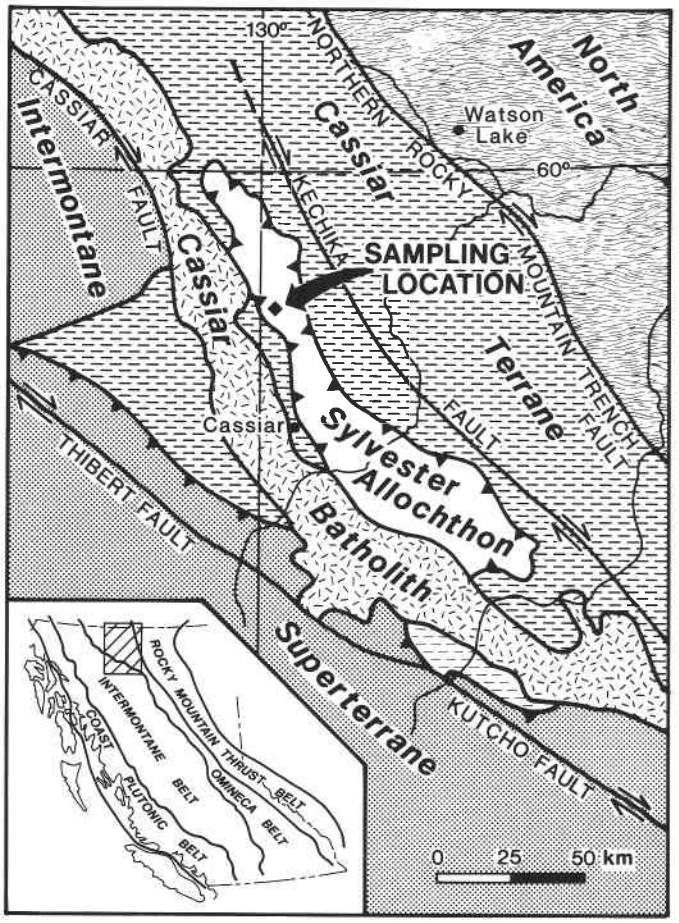


FIG. 1. Regional tectonic setting of the Sylvester Allochthon and the paleomagnetic sampling site.

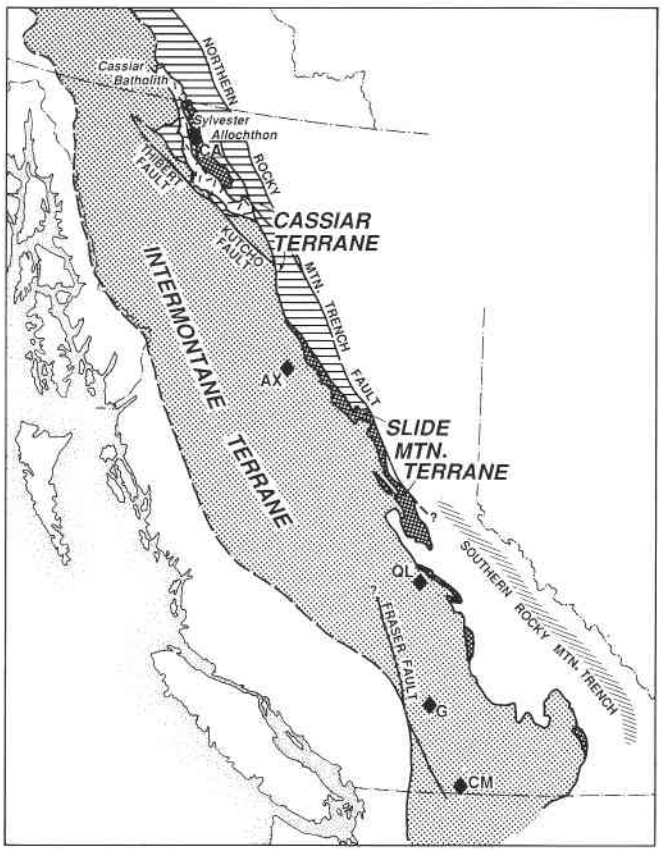


FIG. 2. Distribution of Intermontane Superterrane (stipple), Slide Mountain Terrane (dense stipple), and Cassiar Terrane (horizontal ruling) throughout British Columbia. Also shown are locations of major dextral strike-slip faults and paleomagnetic study localities (AX = Axelgold, G = Guichon, CM = Copper Mountain from Irving *et al.* (1985); QL = Quesnel Lake from Rees *et al.* (1985); CA = Sylvester *Parafusulina* limestone Cassiar Terrane, this paper).

ranges from 10 to 30 km in width. Samples from a number of localities within the batholith have yielded K−Ar isotopic ages of 100−110 Ma, with several concordant muscovite, biotite, and hornblende determinations of about 105 Ma (Gabrielse and Reesor 1974; H. Gabrielse, unpublished data). The Sylvester allochthon was therefore structurally coupled to the Cassiar Terrane by 105 Ma.

A Permian limestone, 125 m thick and approximately 2.5 km in maximum outcrop extent, constitutes one lithotectonic slice of the Sylvester assemblage in the north-central region of the allochthon (Gabrielse 1963; Wolfe 1965). Locally, this limestone is a packstone of distinctive Guadalupian giant *Parafusulina*, known to occur elsewhere only in northern California; near Kettle Falls, Washington; at El Antimonio, Sonora; and in the Word Formation, west Texas (Ross 1969; Ross and Ross 1983). All but the last locality are also in suspect terranes. The Sylvester *Parafusulina* limestone occurs approximately 4 km from the intrusive contact of the Cassiar Batholith. Pinsent and Hirst (1977) defined a contact metamorphic aureole of the Cassiar Batholith by identifying mineral isograds within an ultramafic sliver of the Sylvester Allochthon that lies structurally below the *Parafusulina* limestone and between it and the intrusive contact of the batholith. This clearly indicates significant heating of Sylvester country rock within several kilometres of the batholith. Although neither intruded nor visibly recrystallized, the Sylvester *Parafusulina* limestone was probably also affected by the thermal aureole of the Cassiar Batholith.

The Northern Rocky Mountain Trench Fault defines the eastern margin of the Cassiar Terrane and lies inboard of the terrane, the Sylvester Allochthon, and the Cassiar Batholith (Fig. 1). The Northern Rocky Mountain Trench Fault offsets platformal strata of the Cassiar Terrane from the rest of the North American Paleozoic miogeocline. Geologic evidence indicates 700−900 km of northward translation of the Cassiar Terrane along this major dextral strike-slip fault occurred predominantly from the mid-Cretaceous to the Eocene (Gabrielse 1985), subsequent to intrusion of the Cassiar Batholith. Also, the western contact of the Cassiar Batholith is mylonitized by the Cassiar Fault, probably one of a system of dextral faults related to the Northern Rocky Mountain Trench Fault (Gabrielse 1985).

Outboard of the Cassiar Terrane lies the composite Intermontane Superterrane I (Fig. 2), amalgamated from the Stikine, Cache Creek, and Quesnel terranes prior to accretion to North America in the Middle Jurassic (Monger *et al.* 1982). In southern British Columbia, the Slide Mountain Terrane includes units that are believed to have been amalgamated with the Intermontane Superterrane (Monger *et al.* 1982; Price *et al.* 1985; Klepacki and Wheeler 1985). The Sylvester Allochthon is part of the discontinuous Slide Mountain Terrane. It was also presumably emplaced onto ancestral North

America during collision of the superterrane. However, the allochton has no known root zone either east of or within the Intermontane Superterrane.

Mid-Cretaceous paleomagnetic evidence from the Intermontane Superterrane in southern British Columbia (Fig. 2) suggests it has undergone significant (≥2000 km) post-accretionary northward translation. By implication, this suggests that southern portions of the Sylvester-equivalent Slide Mountain Terrane and the parts of North America to which the superterrane had become welded were displaced as well (Rees *et al.* 1985; Irving *et al.* 1985). Enigmatically in contrast to the Northern Rocky Mountain Trench Fault System, no significant dextral faults are recognized in southern British Columbia between the Intermontane Superterrane and ancestral North America or within the Rocky Mountain Fold and Thrust Belt to the east. The southern continuation of the Northern Rocky Mountain Trench Fault is uncertain. Displacement has not been recognized across the southern part of the Rocky Mountain trench. Price and Carmichael (1986) and Price (1979) suggested distribution of some (about 350 km) offset from the Northern Rocky Mountain Trench Fault into convergence in the fold and thrust belt and transfer of minor (70−100 km) dextral translation to the en échelon Fraser River Fault (Monger 1985) through a zone of ductile extension. However, the Fraser River Fault lies at the western edge of the Intermontane Superterrane, west of many of the study sites that have yielded the discordant paleomagnetic directions.

Paleomagnetism

In an attempt to determine the amount of post-mid-Cretaceous motion of the Cassiar Terrane and the Sylvester Allochthon, we collected eight or more oriented cores at each of 22 sites within the Cassiar Batholith. Unfortunately, the natural remanent magnetization (NRM) of the batholith samples has very low stability and is dominated by viscous components of recent origin. No magnetic cleaning technique was successful in isolating a component of NRM that could reasonably be interpreted as of ancient origin. Therefore, attempts to determine the Cretaceous geomagnetic field direction by paleomagnetic study of the Cassiar Batholith were abandoned.

Although originally collected with hopes of determining a primary late Paleozoic magnetization direction for the Sylvester Allochthon, samples from the Sylvester *Parafusulina* limestone unit that occur near the contact with the Cassiar Batholith appear to have been remagnetized by the batholith and thereby provide a record of the Cretaceous magnetic field direction. Six cores were collected using standard coring techniques at each of 12 sites in limestone (Fig. 1). Cores from each site were spread laterally over several metres of outcrop but generally within a 0.5 m stratigraphic interval. Cores were oriented with Brunton compass and inclinometer. Because of moderate friability, generally only one specimen could be prepared from each core.

NRM was measured with a cryogenic magnetometer, alternating-field (AF) demagnetization was performed with a tumbling specimen demagnetizer, and thermal demagnetization was achieved with a mu-metal shielded furnace in magnetic field of <15 nT in the specimen region. In order to examine the stability of NRM and the coercivity − blocking temperature spectra, progressive AF demagnetization was performed on one sample from each site and progressive thermal demagnetization was done on one additional sample from each

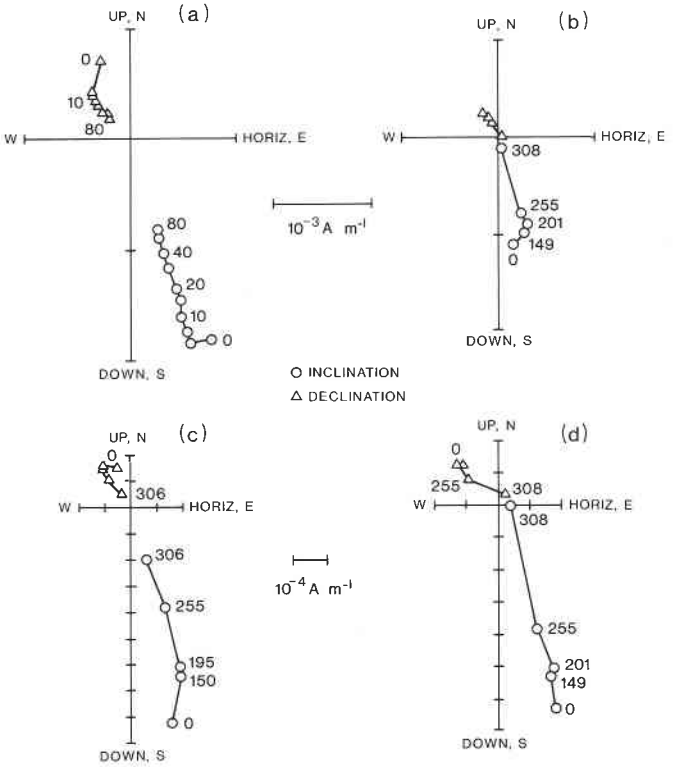


FIG. 3. Vector demagnetization diagrams of progressive demagnetization results. Numbers adjacent to data points are peak demagnetizing field (mT) for sample SY002B (*a*) and temperature (°C) for samples SY002C (*b*), SY007F (*c*), and SY001A (*d*). Upper and lower scales apply to (*a, b*) and (*c, d*), respectively.

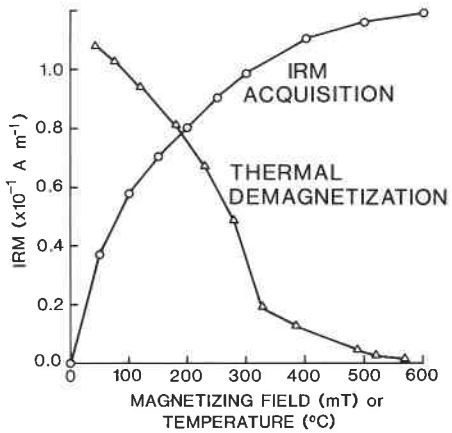


FIG. 4. Acquisition and subsequent thermal demagnetization of IRM.

site. Vector demagnetization diagrams and component analysis (Kirschvink 1980) were used in the analysis.

Typical results of the progressive demagnetization experiments are shown in Fig. 3. The NRM is quite resistant to AF demagnetization (Fig. 3*a*), with almost 50% of the original NRM intensity remaining after demagnetization to 80 mT peak field. However, thermal demagnetization reveals blocking temperatures dominantly in the 150−310°C range. Thermal demagnetization to higher temperatures did not reveal additional components of NRM but caused growth of viscous components of magnetization, probably because of generation

TABLE 1. Paleomagnetic results from remagnetized Sylvester *Parafusulina* limestone for the Cassiar Terrane

Site	I (°)	D (°)	N	R	k	α_{95} (°)	J ($\times 10^{-4}$ A m^{-1})	I' (°)	D' (°)
SY001	70.2	293.2	4	3.99	289.9	5.4	5.4	39.5	265.5
SY002	74.0	311.1	4	3.98	165.6	7.2	9.3	45.9	267.9
SY003	70.2	335.7	6	5.57	11.7	20.4	8.3	50.0	278.8
SY004	81.9	288.4	4	3.85	20.6	20.7	9.8	48.6	255.1
SY005	77.0	309.1	6	5.96	111.9	6.4	13.8	47.5	264.1
SY006	77.5	300.4	4	3.53	6.4	39.4	11.5	49.5	300.8
SY007	74.1	315.8	6	5.98	218.5	4.5	8.8	46.4	307.1
SY008	66.2	358.7	4	4.00	1006.3	2.9	18.1	45.0	329.8
SY009	78.4	357.7	6	5.98	316.1	3.8	21.5	54.4	317.8
SY010	78.2	348.8	4	4.00	620.9	3.7	24.5	71.2	298.2
SY011	80.9	343.6	6	5.96	113.0	6.3	46.6	64.8	272.8
SY012	74.5	359.8	5	4.99	305.9	4.4	55.6	66.3	290.0

NOTES: I, D are site mean inclination and declination (*in situ*); I', D' are site mean inclination and declination (structurally corrected); N is number of sample directions at each site; R is resultant vector length of the N unit vectors; k is estimate of Fisher precision parameter; α_{95} is radius of cone of 95% confidence about site mean direction; J is site mean intensity of the segment of progressive demagnetization that was fit by least squares to determine sample directions.

of a ferrimagnetic mineral during heating. The high coercivity of NRM and blocking temperatures below 320°C strongly suggest that the NRM is carried by pyrrhotite, which has a Curie temperature of 320°C (Schwarz and Vaughan 1972). Acquisition of isothermal remanent magnetism (IRM) and subsequent thermal demagnetization (Fig. 4) show high coercivity for acquisition of IRM and blocking temperatures dominantly below 320°C. Therefore, properties of both the NRM and IRM indicate that pyrrhotite is the carrier of the remanent magnetism.

The progressive thermal demagnetization results (Fig. 3) indicate that the characteristic component of NRM is effectively isolated in the 150−310°C temperature range. However, in some cases (Fig. 3*d*), the NRM vector does not trend directly to the origin of the vector demagnetization plot. The direction of the characteristic component was best determined by least-squares fitting to the four demagnetization steps between 150 and 310°C. Accordingly, all remaining samples were thermally demagnetized in four steps in this temperature interval, and characteristic NRM directions were determined by least-squares fitting to the resulting four vectors for each sample. The characteristic NRM directions were generally well grouped within each site. Typical within-site grouping of characteristic NRM and site mean direction with 95% confidence circle are shown for one site in Fig. 5*a*. The result from one sample (from site SY012) was rejected because of wide deviation between the apparent characteristic NRM for that sample and the characteristic NRM of the other samples at that site. Site mean directions of characteristic NRM and Fisher (1953) statistical parameters are listed in Table 1.

The Sylvester *Parafusulina* limestone is broadly warped across its exposure such that, among the 12 sampling sites, four distinct attitudes were determined, ranging from strike N23°W, dip 35°SW to strike N31°E, dip 28°SE, allowing a fold test to be applied (Graham 1949; Watson and Irving 1957). Site mean directions of characteristic NRM after structural correction and *in situ* are illustrated in Figs. 5*b* and 5*c*, respectively. Averages of site mean directions and of site mean virtual geomagnetic poles (VGP's) along with statistical parameters are listed in Table 2. Comparison of the estimate of the precision parameter (k) before (k = 102.4) and after (k = 21.4) structural correction shows the site mean directional data set fails the fold test at the 99.5% confidence level, clearly indicating that the characteristic NRM was acquired after tilting. We interpret these results as indicating that the characteristic NRM of declination 328.3°, inclination 76.6° is a thermoremanent or thermochemical remanent magnetism acquired by heating and subsequent slow cooling of the Sylvester *Parafusulina* limestone during and soon after intru-

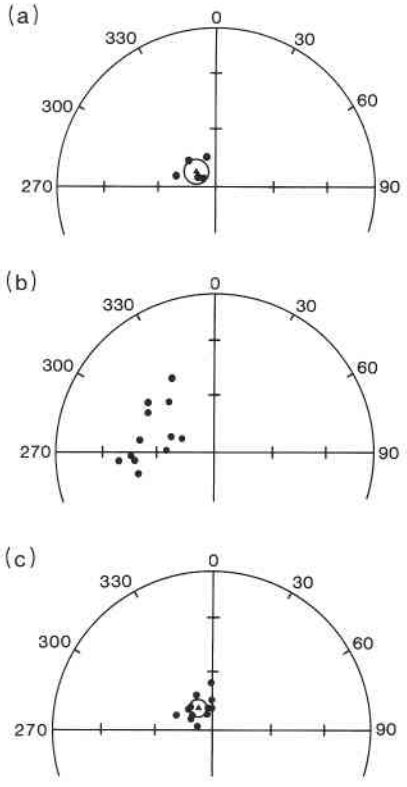


FIG. 5. Equal-area projections of (*a*) *in situ* sample directions and site mean direction for site SY005; (*b*) 12 structurally corrected site mean directions; and (*c*) 12 *in situ* site mean directions with mean of site means and α_{95} confidence circle.

TABLE 2. Average paleomagnetic data

	Average of site mean directions					
	N	R	k	α_{95} (°)	I (°)	D (°)
In situ	12	11.89	102.41	4.3	76.6	328.3
Corrected	12	11.49	21.39	9.6	54.8	286.2

	Average of VGP's					
	N	R	k	α_{95} (°)	Lat. (°N)	Long. (°E)
In situ	12	11.59	27.05	8.5	75.7	171.7
Corrected	12	11.18	13.45	12.3	37.8	147.8

NOTES: "Corrected" indicates data have been structurally correct for local tilt of bedding; N, is number of site means; R, k, and α_{95} are defined in notes for Table 1; Lat., Long. are latitude and longitudes of mean pole positions.

sion of the Cassiar Batholith at 105 Ma. This is consistent with evidence from adjacent Sylvester rocks for a thermal aureole associated with the batholith (Pinsent and Hirst 1977). Time integration of the geomagnetic field during this process is thought to be responsible for the observed VGP angular dispersion of 15.6°, which is somewhat less than the approximately 20° expected from complete sampling of geomagnetic secular variation at a paleolatitude of 70° (Merrill and McElhinny 1983).

No correction for regional tilt was applied to these data, as no direct indicators of paleohorizontal at the time of remagnetization are available. However, the base of the Sylvester Allochthon is at present an essentially unbroken, planar, and horizontal fault over the hundreds square kilometres of its outcrop area (Harms 1984). This seems to preclude significant post-emplacement or post-intrusion tilt of broad crustal blocks. Note also that most structures in the area are subparallel and impart a large-scale west-northwest-trending structural grain to the region, which is followed by the major transcurrent faults. As the declination of NRM determined for the Cassiar Terrane from the Sylvester *Parafusulina* limestone parallels this structural grain, undetected post-magnetization tilting about axes in the structural grain would produce minimal errors in inclination of the magnetization. Thus the indirect geological evidence strongly indicates that no significant post-magnetization tilting has occurred.

Discussion and conclusions

In selection of the appropriate mid-Cretaceous cratonic reference pole for comparison with our pole determined from the Cretaceous remagnetization of the Sylvester *Parafusulina* limestone, we follow the analysis of Frei (1986). The mid-Cretaceous reference pole is taken to be the mean of the paleomagnetic poles from (*i*) Niobrara Formation (Shive and Frerichs 1974); (*ii*) alkalic rocks of Arkansas (Scharon and Hsu 1969); and (*iii*) Isachsen diabase (Larochelle and Black 1963); Larochelle *et al.* 1965). These poles were determined from rocks with ages between approximately 90 and 120 Ma (within 15 Ma of the 105 Ma age of the Cassiar Batholith). Also, a reasonable number (≥ 10) of separate beds or igneous units were sampled in each study, thus affording reasonable averaging of geomagnetic secular variation. In addition, at least some magnetic cleaning was undertaken, and all three of these mid-Cretaceous units are situated within the tectonically stable continental interior. The mean of these poles (latitude = 66.4°N, longitude = 187.4°E, A_5 = 3.6°) can be used as an accurate reference pole for the entire mid-Cretaceous because of insignificant apparent polar wander from North America during this time. The quite small error oval is the result of tight clustering of the three poles and may overestimate the accuracy with which the mid-Cretaceous reference pole is known. However, any reasonable approach to defining a mid-Cretaceous reference pole yields results that are very similar. For example, using a sliding time-window technique, Irving and Irving (1982) determined mean poles for 90, 100, and 110 Ma that were all within 2° of the mid-Cretaceous reference poles used here. It is thus clear that the mid-Cretaceous is a time interval for which the cratonic reference pole is unusually well determined.

Calculation of concordance–discordance values and uncertainties follows the techniques of Beck (1980), Demarest (1983), and Beck *et al.* (1986). Results of calculations done both in pole space and in direction space are listed in Table 3. Tectonic displacement is more clearly interpretable in terms of poleward translation (p) calculated in pole space than is inclination shallowing (F) calculated in direction space.

Using 95% confidence limits, we find modest but statistically significant post-105 Ma clockwise vertical axis rotation (23.9 ± 18.1°). However, no paleomagnetically discernible post-105 Ma poleward translation ($p \pm \Delta p = 5.3 \pm 9.2°$) has affected the Cassiar Terrane and overlying Sylvester Allochthon. The reference pole and the pole determined from the Cretaceous remagnetization of the Sylvester *Parafusulina* limestone for the Cassiar Terrane are illustrated in Fig. 6, as are the rotation and poleward translation angles with the attendant 95% confidence limits. As the Cassiar Terrane lies just outboard of a major dextral transcurrent fault, the Northern Rocky Mountain Trench Fault, our observed 24° clockwise rotation of the terrane can be expected, as shown by well-documented rotations outboard of the San Andreas Fault (e.g., Luyendyk *et al.* 1985). Perhaps the only surprising aspect of this observation is the modest amount of rotation.

The absence of significant poleward translation of the Cassiar Terrane (concordant paleolatitude) is a provocative result. Geological evidence indicates displacements of 700–900 km on the Northern Rocky Mountain Trench Fault. The calculated poleward translation of 5.3°, although not significant at the 95% confidence level, equals on the order of 600 km of north-northwesterly transport toward the Creta-

TABLE 3. Results of rotation and translation calculations

Comparison of poles:	$R \pm \Delta R = 23.9 \pm 18.1°$
	$p \pm \Delta p = 5.3 \pm 9.2°$
Comparison of directions:	$R \pm \Delta R = 20.4 \pm 17.3°$
	$F \pm \Delta F = 3.0 \pm 3.8°$

NOTES: Reference pole is mean mid-Cretaceous pole for cratonic North America (66.4°N, 187.4°E, $A_{95} = 3.6°$, see text for discussion and references); Cassiar Terrane pole is mean of *in situ* site mean VGP's from the present study (75.7°N, 171.7°E, $\alpha_{95} = 8.5°$, see text for discussion); listings under "Comparison of poles" are clockwise rotation (R) with 95% confidence limits (ΔR) and poleward translation (p) with 95% confidence limits (Δp) from calculations in pole space. Listings under "Comparison of directions" are results of similar calculations in direction space.

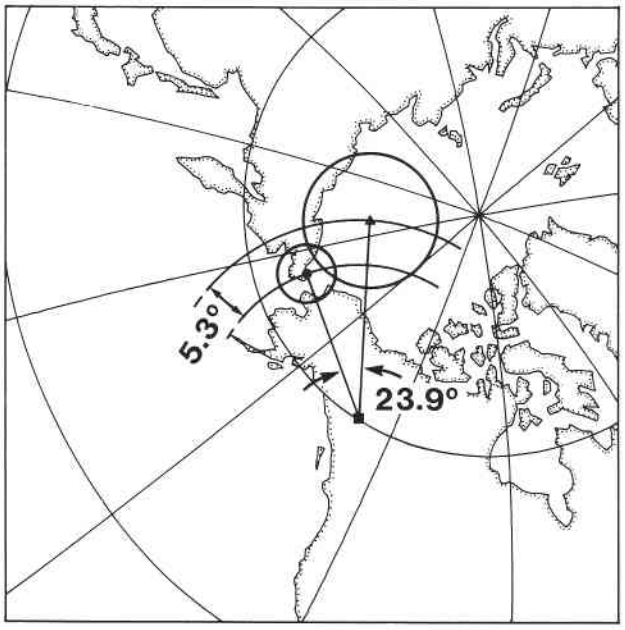


FIG. 6. Comparison of paleomagnetic pole positions. Pole from Cassiar Platform is shown by the triangle with its circle of 95% confidence. Reference mid-Cretaceous pole for cratonic North America is represented by the circle. Sampling location is represented by the square. Apparent rotation and poleward translation are also shown.

ceous reference pole. Thus the paleomagnetic data from the Sylvester *Parafusulina* limestone are consistent with the geological evidence for magnitude of transport of the Cassiar Terrane on the Northern Rocky Mountain Trench Fault during the Late Cretaceous and Tertiary.

These results are, however, in contrast to Cretaceous paleomagnetic results from similar tectonic settings in central and southern British Columbia. Irving *et al.* (1985) and Rees *et al.* (1985) reported paleomagnetic directions from post-accretionary Cretaceous batholiths intruded into the Intermontane Superterrane and from regions of the superterrane remagnetized by those batholiths. These directions are consistently rotated clockwise (by about 70°) and are shallow (by about 20°) with respect to expected directions. The paleolatitudes determined from these magnetizations appear to require as much as 2500 km of post-mid-Cretaceous northward movement of the Intermontane Superterrane, Slide Mountain Terrane, *and* inboard parts of North America to which the superterrane had already been sutured, despite the absence of recognized structures on which to accommodate this translation. Post-magnetization tilting of about 30° toward the southwest could also account for the observations (Rees *et al.* 1985), but this explanation requires an unlikely consistent tilting over central and southern British Columbia as well as northern Washington.

If our interpretations of the paleomagnetic data are correct, no paleomagnetically significant post-mid-Cretaceous latitudinal motion of the Cassiar Terrane is required, beyond the 700–900 km indicated by geological evidence. These results do not solve the apparent conflict between Cretaceous paleomagnetic results from central and southern British Columbia, which seem to require large post-mid-Cretaceous latitudinal transport inboard of the Intermontane Superterrane, and the geologic evidence that is difficult to reconcile with such motion. Nor do they necessarily limit large northward translations in southern British Columbia. However, the present data do preclude such displacement from connecting with structures that lie east of the Cassiar Terrane and are thus in conflict with the model of Chamberlain and Lambert (1985).

The current paleomagnetic results are another piece of a puzzle that is yet to be satisfactorily reconstructed. We agree with Rees *et al.* (1985) that "more work must be done before the enigma is solved."

Acknowledgments

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